

TASK 3 — NETWORK TRAFFIC ANALYSIS

Wireshark + Kali Linux | Practical Cybersecurity Lab Report

1. Objective

Capture and analyze network traffic in a controlled Kali Linux lab using Wireshark, identify important protocols, inspect packet fields, and document findings from a SOC analyst perspective.

2. Environment & Tools

- Kali Linux running in Oracle VirtualBox.
- Wireshark packet analyzer.
- curl for independent HTTP/HTTPS validation.
- Network interface: eth0.

3. Activities Performed

- Started a live packet capture on eth0.
- Used the tcp display filter to isolate TCP/HTTPS traffic.
- Used the dns display filter to isolate DNS queries and responses.
- Inspected TCP SYN/ACK and FIN/ACK packets and TLS 1.3 sessions.
- Inspected DNS A, AAAA and PTR records and Mozilla/Firefox service lookups.
- Validated web connectivity using curl and observed HTTP response codes.

4. Key Findings

- DNS resolution traffic was observed between the local host and DNS service.
- HTTPS communication used TCP port 443 and TLS 1.3.
- TCP connection establishment and orderly termination were visible.
- ICMP echo request/reply traffic demonstrated connectivity.
- ARP and IPv6 control traffic were also present.
- curl validation showed successful HTTP 200 and expected HTTP 301 redirect behavior.

5. SOC Interpretation

The captured traffic is consistent with normal web and network activity. The screenshots alone do not establish compromise. A SOC analyst should correlate packet evidence with authentication logs, endpoint telemetry, DNS reputation, destination context, timing, volume and other indicators before declaring an event malicious.

6. Skills Demonstrated

Wireshark • Packet Analysis • DNS Analysis • TCP/IP • TLS • Network Troubleshooting • Evidence Collection • SOC Investigation • Security Reporting

Evidence Screenshots

Figure 1. Live Wireshark capture with TCP-filtered HTTPS/TLS traffic on port 443.

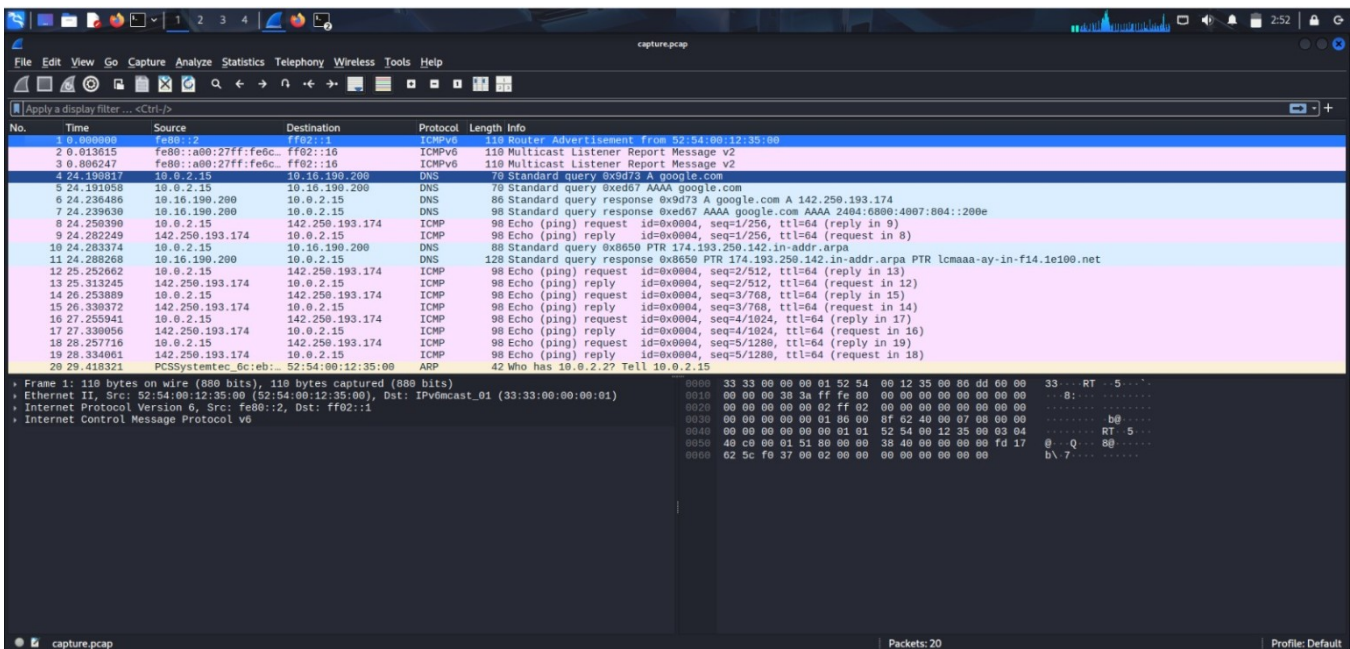


Figure 2. TCP/TLS 1.3 session showing acknowledgements, application data and connection termination.

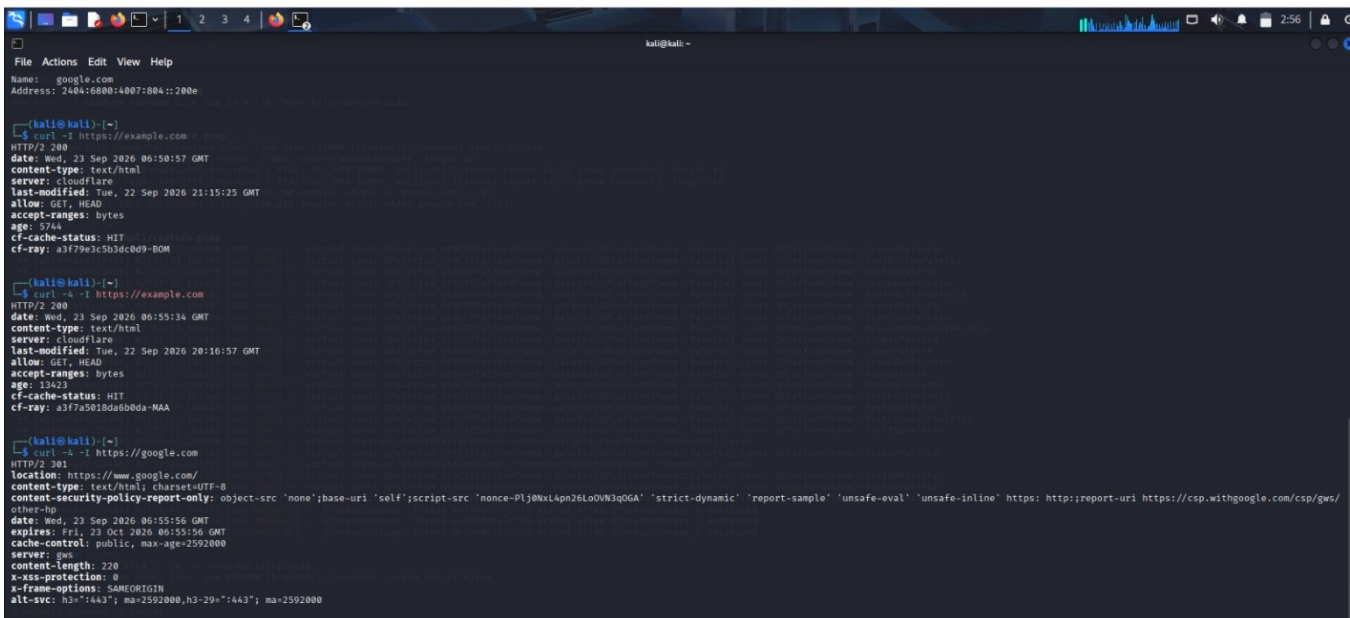


Figure 3. Mixed capture showing TCP, DNS and ICMP traffic, including ping request/reply.

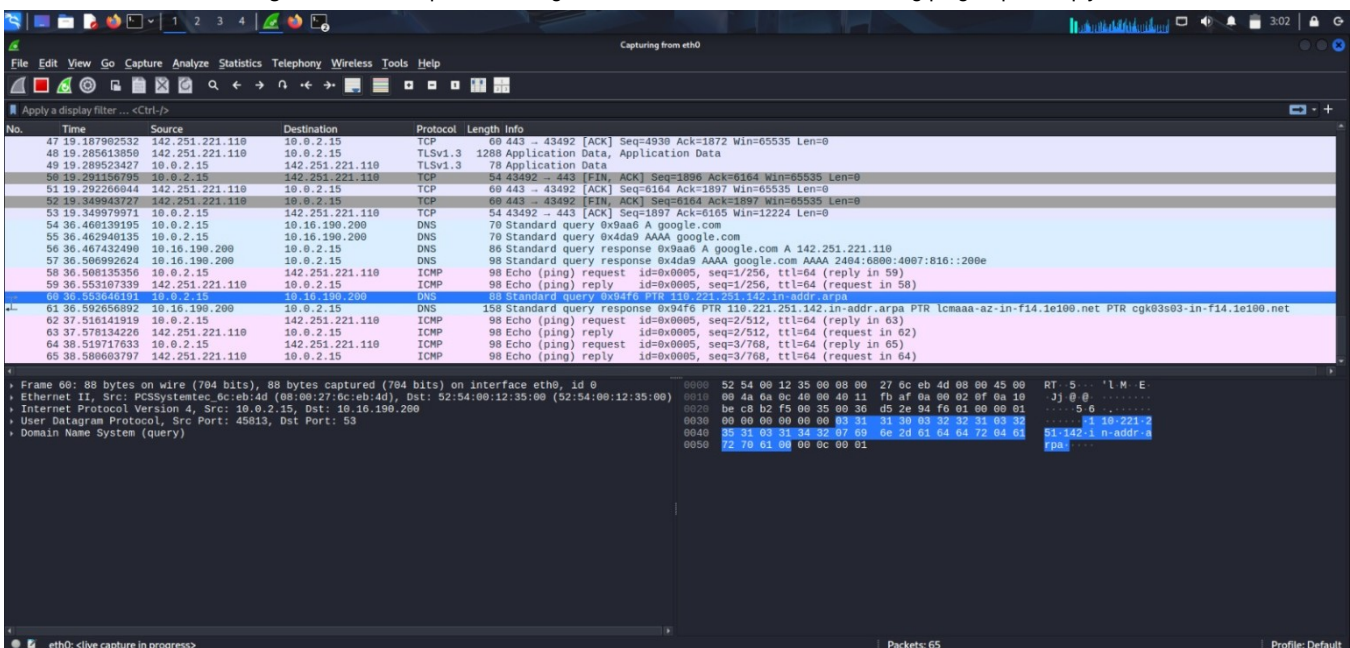


Figure 4. DNS-filtered capture showing A/AAAA/PTR lookups and Mozilla/Firefox service resolution.

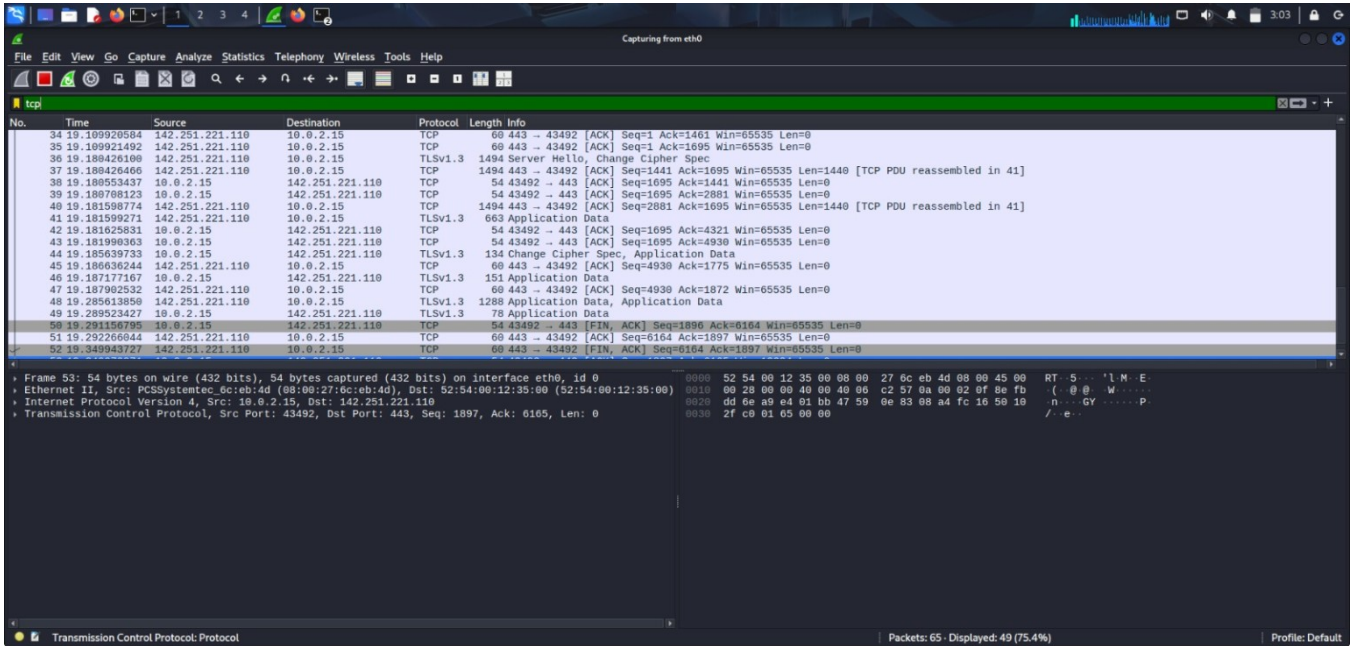


Figure 5. curl output validating HTTP response behavior for example.com and google.com.

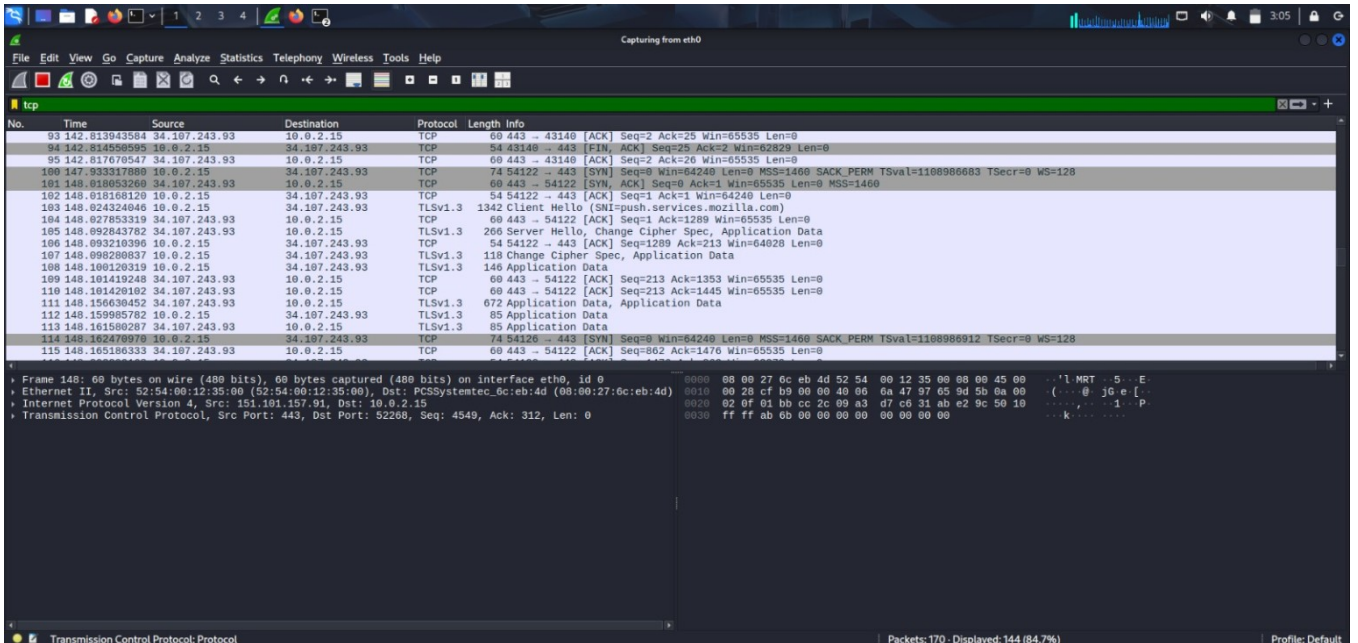


Figure 6. DNS response inspection showing Firefox/Mozilla service-domain resolution details.

